September 14 Classwork : Runner.java and Radius.java:

```java
public class Radius {
   //radius class with methods that prints area, circumference and volume of a specifc
radius(parameter)

   private double pi = 3.14;


   // Calculates the area of a circle from the given radius
   public void printArea(double radius){
       double area = pi * Math.pow(radius,2);
       System.out.println("Area: " + area);
   }
   // Calculates the circumference of a circle from the given radius
   public void printCir(double radius){
       double circumference = 2 * pi * radius;
       System.out.println("Circumference: " + circumference);
   }
   // Calculates the volume of a cone from the given radius and height
   public void printConeVol(double radius,double height){
       double volume = (1.0/3)* pi * height * Math.pow(radius,3);
       System.out.println("Cone Volume: " + volume);
   }

}
```

Runner:

```java


// Creates a Radius object and prints its area, circumference, and cone volume
public class Runner {

   public static void main(String[] args) {
       Radius radius = new Radius();

       radius.printArea(10);
       radius.printCir(10);
       radius.printConeVol(10, 5);



   }
```

```
}
```

Output:

```
yuvan@MacBook-Air-55 sept14classwork % java Runner
yuvan@MacBook-Air-55 sept14classwork % javac Runner.java
yuvan@MacBook-Air-55 sept14classwork % java Runner
Area: 314.0
Circumference: 62.800000000000004
Cone Volume: 5233.333333333333
yuvan@MacBook-Air-55 sept14classwork %
```

September 14/15 or September 14/16 Warm-Up: Animal.java and Runner.java (code and output):

Animal:

```
public class Animal {
   private String animalType;
   private int animalAge;

   // Sets the animal's type and age instance variables from the parameters
   public void setVariables(String type, int age){
       animalType  = type;
       animalAge = age;

   }
   // Prints the animal's current type and age
   public void printInfo(){
       System.out.println("Animal type is " + animalType + " age is " + animalAge);

   }
   //code for the animal object, methods and instance variables
}
```

Runner:

```
import java.util.Scanner;

// Reads animal type and age from user input, then prints info for two animals
public class Runner {

   public static void main(String[] args) {
```

```
        String type;
        int age;
        Scanner scanner = new Scanner(System.in);
        Animal aObj1 = new Animal();
        Animal aObj2 = new Animal();
        System.out.print("What is the type of animal? ");
        type = scanner.nextLine();
        System.out.print("What is the animal's age? ");
        age = scanner.nextInt();
        aObj1.setVariables(type, age);
        aObj1.printInfo();
        aObj2.setVariables("Horse", 7);
        aObj2.printInfo();
        scanner.close();
    //code for the runner that calls methods on the animal class}
}
}
```

Output:

```
yuvan@MacBook-Air-55 sept14andsept15warmup % javac Runner.java
yuvan@MacBook-Air-55 sept14andsept15warmup % java Runner
What is the type of animal? raymond
What is the animal's age? 14
Animal type is raymond age is 14
Animal type is Horse age is 7
yuvan@MacBook-Air-55 sept14andsept15warmup %
```

September 14/16 Classwork #1 (from dropbox)

```
public class Animal {
    private String animalType, animalName = "no name";
    private int age;

    //default constructor that initializes animalType and age
    public Animal() {
        animalType = "dog";
        age = 5;
    }
    // Constructor that sets type, name, and age all at once from parameters
    public Animal(String animalT, String name, int animalAge){
            animalType = animalT;
            animalName = name;
            age = animalAge;
```

```
    }

    //initialization constructor that accepts parameters to initialize animalType and
age
    public Animal(String animalT, int animalAge) {
        animalType = animalT;
        age = animalAge;
    }

    //set animalName to the passed in parameter
    public void setName(String name) {
        animalName = name;
    }

    // Print the animal type and age
    public void printInfo() {
        System.out.println("The animal's type is " + animalType + ".");
        System.out.println("The animal's age is " + age + ".");
        System.out.println("The animal's name is " + animalName + ".\n");
    }
}
```

Runner:

```
public class Runner {
    // Creates several Animal objects using different constructors and prints their
info



    public static void main(String args[]) {



        Animal animal1 = new Animal();
                //Comment: instantiating a object of class animal called animal 2 with
the default constructor
        Animal animal2 = new Animal("cat", 7);
                //Comment:  instantiating a object of class animal called animal 2 with
the parameters animalType: cat and animalAge:7
```

```
        Animal animal3 = new Animal("goat", 2);

        Animal animal4 = new Animal("human", "arya", 15);

                //Comment: Instantiating an object of class animal that is called
animal3 with animal type set as goat and the age set to 2
        animal1.printInfo();



                //Comment: Printing the type, name and age of animal 1 (name is not
applicable). It will not print anything
        animal2.setName("Roco");
        animal2.printInfo();



                //Comment: printing the type, name and age of animal 2. The name is
roco, it is a cat and its age is 7
        animal3.setName("Fluffy");
        animal3.printInfo();
                //Comment: sets the name of animal 3 to fluffy. then it prints all of
its properties, which its name (fluffy), type(goat) and age(2)
        animal4.printInfo();
    }
}
```

Output:

```
yuvan@MacBook-Air-55 week6 % cd constructorExample
yuvan@MacBook-Air-55 constructorExample % javac Runner.java
yuvan@MacBook-Air-55 constructorExample % java Runner
The animal's type is dog.
The animal's age is 5.
The animal's name is no name.

The animal's type is cat.
The animal's age is 7.
The animal's name is Roco.

The animal's type is goat.
The animal's age is 2.
The animal's name is Fluffy.

The animal's type is human.
The animal's age is 15.
The animal's name is arya.
```

September 14/16 Classwork #2: Triangle.java, Rectangle.java and Runner.java (code and output)

Rectangle:

```java
// Represents a rectangle with a length and width and can print its area
public class Rectangle{
   private int length, width;

   // Default constructor that sets length and width to zero
   public Rectangle(){
       length = 0;
       width = 0;

   }
   // Constructor that sets length and width from the given parameters
   public Rectangle(int length, int width){
       this.length = length;
       this.width = width;
   }
   // Calculates and prints the rectangle's area
   public void printArea(){
       int area = length * width;
       System.out.println("Area is " + area);
   }



}
```

Triangle.java

```java
// Represents a triangle with a base and height and can print its area
public class Triangle{

   private int base;
   private int height;

   // Default constructor that sets base and height to zero
   public Triangle(){
       base = 0;
       height = 0;
   }
   // Constructor that sets base and height from the given parameters
   public Triangle(int base, int height){
```

```
        this.base = base;
        this.height = height;

    }
    // Calculates and prints the triangle's area
    public void printArea(){
        double area = (base * height) / 2.0;
        System.out.println("Area is " + area);
    }




}
```

Runner:

```
// Creates Rectangle and Triangle objects and prints each of their areas
public class Runner{

    public static void main(String[] args) {
        Rectangle rec1 = new Rectangle();
        rec1.printArea();
        Rectangle rec2 = new Rectangle(5,3);
        rec2.printArea();
        Triangle triangle1 = new Triangle();
        triangle1.printArea();
        Triangle triangle2 = new Triangle(5,3);
        triangle2.printArea();
        //All object insantiations and calling methods for triangle objects



    }
}
```

Output:

```
yuvan@MacBook-Air-55 sept15classwork %
yuvan@MacBook-Air-55 sept15classwork % javac Runner.java
yuvan@MacBook-Air-55 sept15classwork % java Runner
Area is 0
Area is 15
Area is 0.0
Area is 7.5
yuvan@MacBook-Air-55 sept15classwork %
```

5) September 17/18 Classwork: Gui.java or Scenery.java (No Runner) - code and output

GuiEX.java:

```
import javax.swing.JPanel;

import java.awt.Color;

import java.awt.Graphics;

import java.awt.Dimension;




// Class that sets up the JPanel and draws on the panel

public class GuiEx extends JPanel {

   private int x, y;



       // Constructor that initializes instance variables, etc.

   public GuiEx() {

       setFocusable(true);  // set to true by default, so really don't need this.

       setLayout(null);  /* Allows us to control the positioning of components on

                            the panel. If not set to null, BorderLayout is the default
layout. */

       x = 400;

       y = 200;

   }




       // Sets up the dimensions of the JPanel

   @Override

   public Dimension getPreferredSize() {

       //Sets the size of the panel
```

```
        return new Dimension(800,600);
    }



        // Contains calls to methods that draw on the JPanel
    @Override
    public void paintComponent(Graphics g) {
        super.paintComponent(g);           // DO NOT REMOVE THIS. You MUST call the
superclass paintComponent(Graphics) to clear the background.


                drawTree(g);


    }




        private void drawTree(Graphics g){
            Color brown = new Color(150, 75, 0);
            g.setColor(brown);
            g.fillRect(x+75, y+75, 50, 200);


            Color green = new Color(0,255,0);
            g.setColor(green);
            g.fillOval(x,y, 200, 200);
            //Makes a tree stump that is a brown rectangle and a green oval on top of
it to represent leaves
```

```
        }



}
```

Output:

